

Debugging Report

Failure Simulation and Analysis

1) Locator Failure: Simulate an error caused by an incorrect locator.

Steps to Simulate:

1. Write a standard Playwright Code
2. Provide a non-existent or misspelt locator string(eg #tbButton)
3. Execute the test

Actual Implementation:

Change newTabButton: '#tabButton' in selectors.ts to #tbButton

Execute the following:

`npx playwright test browserWindows.spec.ts`

Failure Log:

```
Error: locator.click: Test timeout of 30000ms exceeded.  
Call log:  
  - waiting for locator('#tbButton')  
  
at ..\pages\BrowserWindowsPage.ts:24
```

HTML Report of that test:

All 12

✓ Passed 3

✗ Failed 9

Flaky 0

Skipped 0

🔄

next

TC_001: Verify New Tab button opens a new tab

browserWindows.spec.ts:4

chromium

Run

Errors

Test timeout of 30000ms exceeded.

Copy prompt

Error: locator.click: Test timeout of 30000ms exceeded.

Call log:

- waiting for locator('#tbButton')

at ../pages/BrowserWindowsPage.ts:24

```

22 |
23 |     async openNewTab(): Promise<void>{
> 24 |         await this.newTabButton.click();
    |                                     ^
25 |     }
26 |
27 |     async openNewWindow(): Promise<void>{
    at BrowserWindowsPage.openNewTab (C:\Users\dnyanesh.panchal\Desktop\AutomationTask_2\pages\BrowserWindowsPage.ts:24:
    at C:\Users\dnyanesh.panchal\Desktop\AutomationTask_2\tests\browserWindows.spec.ts:8:28

```

Test Steps

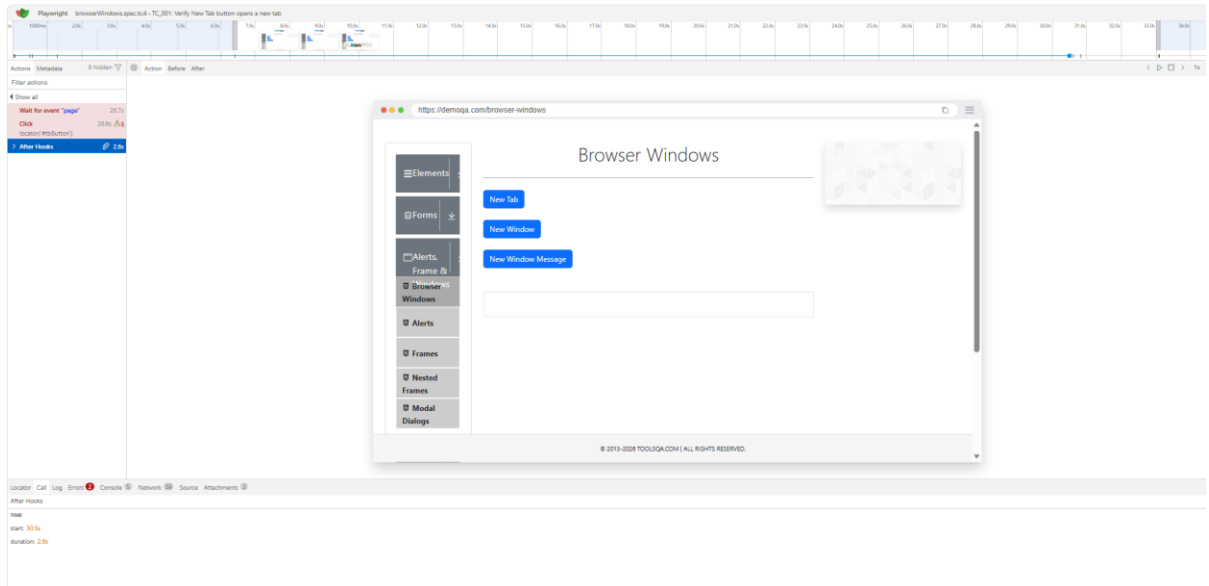
> ✓ Before Hooks	1.3s
> ✓ Navigate to "/browser-windows" --- ../pages/BrowserWindowsPage.ts:20	5.1s
> ✗ Wait for event "page" --- browserWindows.spec.ts:7	26.7s
> ✗ Click locator('#tbButton') --- ../pages/BrowserWindowsPage.ts:24	26.6s
> ✓ After Hooks	2.6s
> ✓ Worker Cleanup	63ms

Screenshots

📷

screenshot

TraceViewer Evidence:



Root Cause:

The locator did not match any element present in the DOM causing Playwright to continuously search for the element until the timeout period was reached.

Resolution:

Verified locator using browser DevTools and restored the correct selector

2) Assertion Failure: Simulate a validation error due to incorrect expected value

Steps to simulate:

In BrowserWindowsPage.ts, change:

```
async verifySamplePageContent(childPage: Page):Promise<void>{  
    await expect(  
        childPage.locator(selectors.sampleHeading)  
        .toHaveText(messages.samplePageText)  
    )  
}
```

To

```
async verifySamplePageContent(childPage: Page):Promise<void>{  
    await expect(  
        childPage.locator(selectors.sampleHeading)  
        .toHaveText('I am Batman')  
    )  
}
```

Execute:

`npx playwright test browserWindows.spec.ts`

Failure Log:

```

PS C:\Users\dnyanesh.panchal\Desktop\AutomationTask_2> npx playwright test browserWindows.spec.ts

Running 12 tests using 4 workers
1) [firefox] › tests\browserWindows.spec.ts:37:5 › TC_004 Verify New Window Message functionality.

Test timeout of 30000ms exceeded.

Error: page.goto: Test timeout of 30000ms exceeded.
Call log:
- navigating to "https://demoqa.com/browser-windows", waiting until "load"

```

HTML Report of that test:

Search tests

All 12 ✓ Passed 11 ✗ Failed 1 Flaky 0 Skipped 0

TC_004 Verify New Window Message functionality.

browserWindows.spec.ts:37

View Trace | 31.1s

Firefox

Run

Errors

Test timeout of 30000ms exceeded.

Error: page.goto: Test timeout of 30000ms exceeded.
Call log:
- navigating to "https://demoqa.com/browser-windows", waiting until "load"

at .\pages\BrowserWindowsPage.ts:20

```

18 |   }
19 |   async goto(): Promise<void> {
20 |     await this.page.goto(routes.browserWindows);
21 |   }
22 |
23 |   async openNewTab(): Promise<void> {
    at BrowserWindowsPage.goto (C:\Users\dnyanesh.panchal\Desktop\AutomationTask_2\pages\BrowserWindowsPage.ts:20:25)
    at C:\Users\dnyanesh.panchal\Desktop\AutomationTask_2\tests\browserWindows.spec.ts:38:30

```

Test Steps

Filter steps

Step	Duration
✓ Before Hooks	3.7s
✗ Navigate to "/browser-windows"/pages/BrowserWindowsPage.ts:20	29.2s
✓ After Hooks	1.1s
✓ Worker Cleanup	674ms

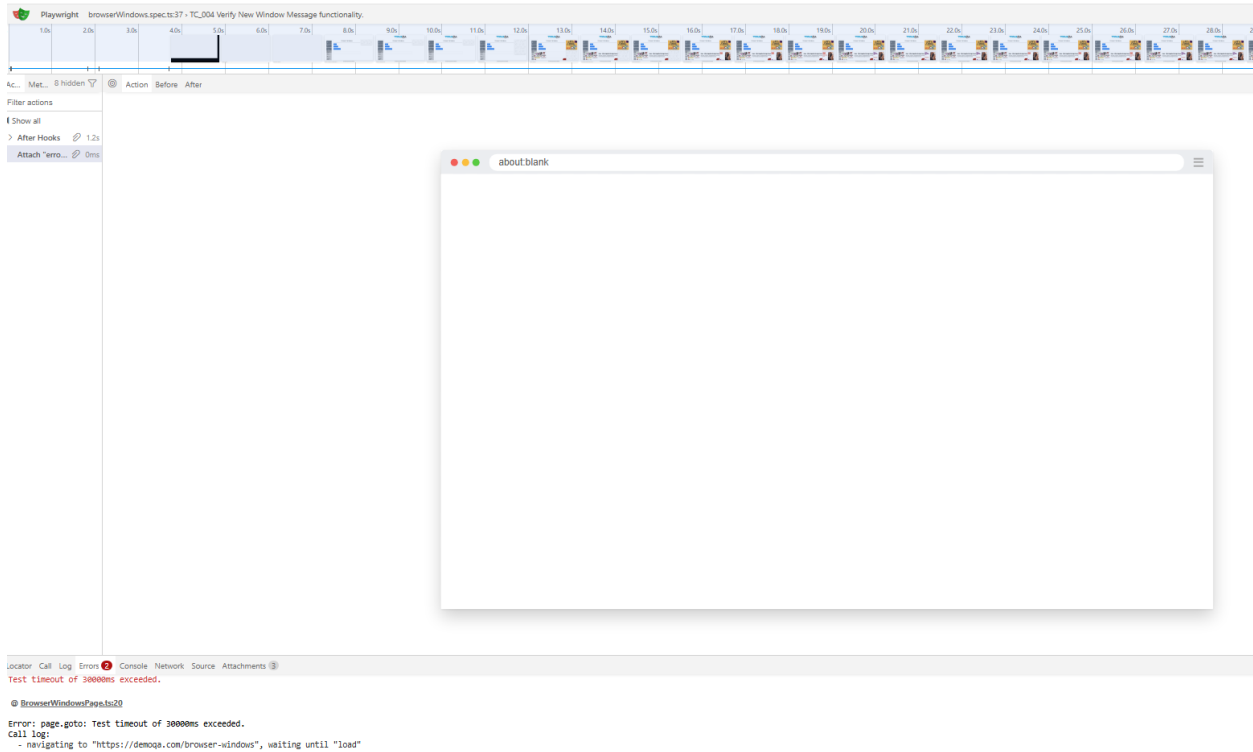
Screenshots



Traces



TraceViewer Evidence:



Root cause:

Expected value did not match actual application content

Resolution:

Updated expected value should match application behavior using DevTools.

3) Network Mocking Failure: Simulate incorrect API route interception

Steps to simulate:

In networkMocking.spec.ts, change and create a new similar test file

await page.route(

'**//BookStore/v1/Books',

To

await page.route(

'**//BookStore/v1/InvalidBooks',

Run:

npx playwright test networkMocking.spec.ts

(Main changes are: Modified the network interception logic to intentionally throw an exception inside the route handler)

Failure Log:

```
PS C:\Users\dnyanesh.panchal\Desktop\AutomationTask_2> npx playwright test networkFailure.spec.ts

Running 3 tests using 3 workers
1) [chromium] › tests\networkFailure.spec.ts:5:5 › Network Mock Failure Simulation

Error: expect(locator).toBeVisible() failed

Locator: getByText('Playwright Mastery')
Expected: visible
Timeout: 5000ms
Error: element(s) not found
```

HTML Report of the test:

Network Mock Failure Simulation

networkFailure.spec.ts:5

View Trace | 11.6s

chromium

Run

Errors

Error: expect(locator).toBeVisible() failed
Locator: getByText('Playwright Mastery')
Expected: visible
Timeout: 5000ms
Error: element(s) not found

Copy prompt

Call log:

- Expect "toBeVisible" with timeout 5000ms
- waiting for getByText('Playwright Mastery')

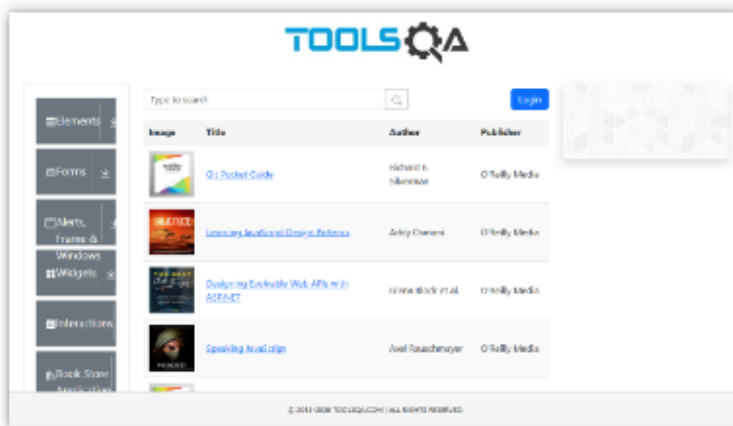
```
51 |     page.getByText('Playwright Mastery')  
52 |  
> 53 |     ).toBeVisible();  
    |       ^  
54 |  
55 |  
56 |  
at C:\Users\dnyanesh.panchal\Desktop\AutomationTask_2\tests\networkFailure.spec.ts:53:5
```

Test Steps

Filter steps

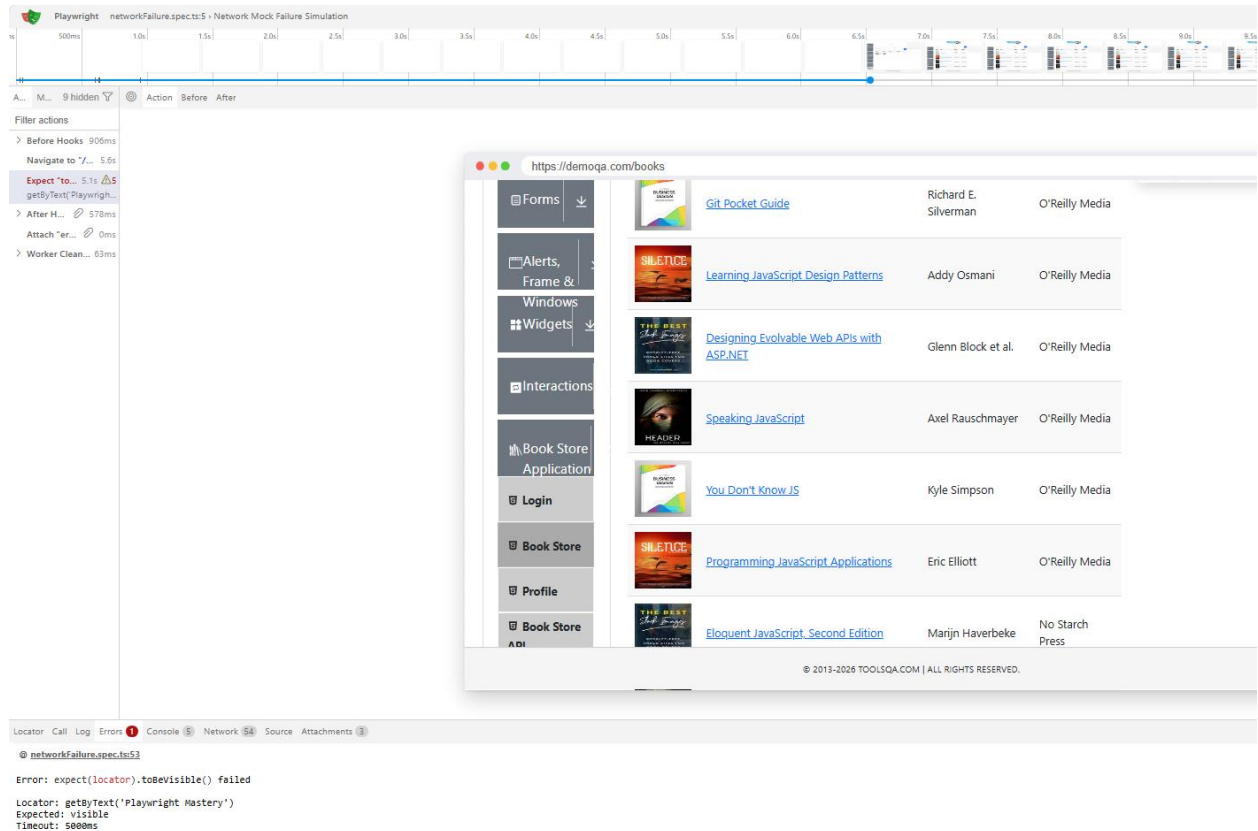
> ✓ Before Hooks	906ms
> ✓ Navigate to "/books" — networkFailure.spec.ts:41	5.6s
> ✗ Expect "toBeVisible" getByText('Playwright Mastery') — networkFailure.spec.ts:53	5.1s
> ✓ After Hooks	578ms
> ✓ Worker Cleanup	63ms

Screenshots



screenshot

TraceViewer Evidence:



Root Cause:

The root handler never intercepted the API request because the URL pattern was incorrect.

Resolution:

Verified Endpoint using browser Network tab and restored correct route matcher.

4) Mock Response Structure Failure: Simulate failure caused by malformed mock response

Steps to Simulate:

1. Create a file : mockResponseFailure.spec.ts
2. Add the following:

```
import { test, expect } from '@playwright/test';

test('Mock Response Failure Simulation', async ({ page }) => {

  await page.route(

    '**/BookStore/v1/Books',

    async route => {

      await route.abort();

    }

  );

  await page.goto('https://demoqa.com/books');

  await expect(

    page.getByText('Playwright Mastery')

  ).toBeVisible();

});
```

3. Run
npx playwright test mockResponseFailure.spec.ts

Failure Log:

```
Running 3 tests using 3 workers
1) [webkit] > tests/mockResponseFailure.spec.ts:5:5 > Mock Response Failure Simulation

Error: expect(locator).toBeVisible() failed

Locator: getByText('Playwright Mastery')
Expected: visible
Timeout: 5000ms
Error: element(s) not found

Call log:
  - expect "toBeVisible" with timeout 5000ms
    - waiting for getByText('Playwright Mastery')
```

HTML Report of the test:

Q Search tests ALL Passed Failed Ruby Skipped and +

Mock Response Failure Simulation

mockResponseFailure.spec.ts View Trace 96s

chromium

Run

Errors

```

Error: expect(locator).toBeVisible() failed
Locator: getByText('Playwright Mastery')
Expected: visible
Timeout: 5000ms
Error: element(s) not found

Call log:
  - expect "toBeVisible" with timeout 5000ms
    - waiting for getByText('Playwright Mastery')

29   page.getByText('Playwright Mastery')
30   ).toBeVisible();
31   ^
32
33
34 at C:\Users\dnyanedh.panchal\Desktop\automation\tests\mockResponseFailure.spec.ts:31:5
        
```

[Copy prompt](#)

Test Steps

Filter steps

> ✓ Before Hook	560ms
> ✓ Navigate to "/book" — mockResponseFailure.spec.ts:21	3.3s
> ✗ Expect "toBeVisible" getByText("Playwright Mastery") — mockResponseFailure.spec.ts:31	5.1s
> ✓ After Hook	960ms
> ✓ Worker Cleanup	930ms

Screenshots



[Download](#)

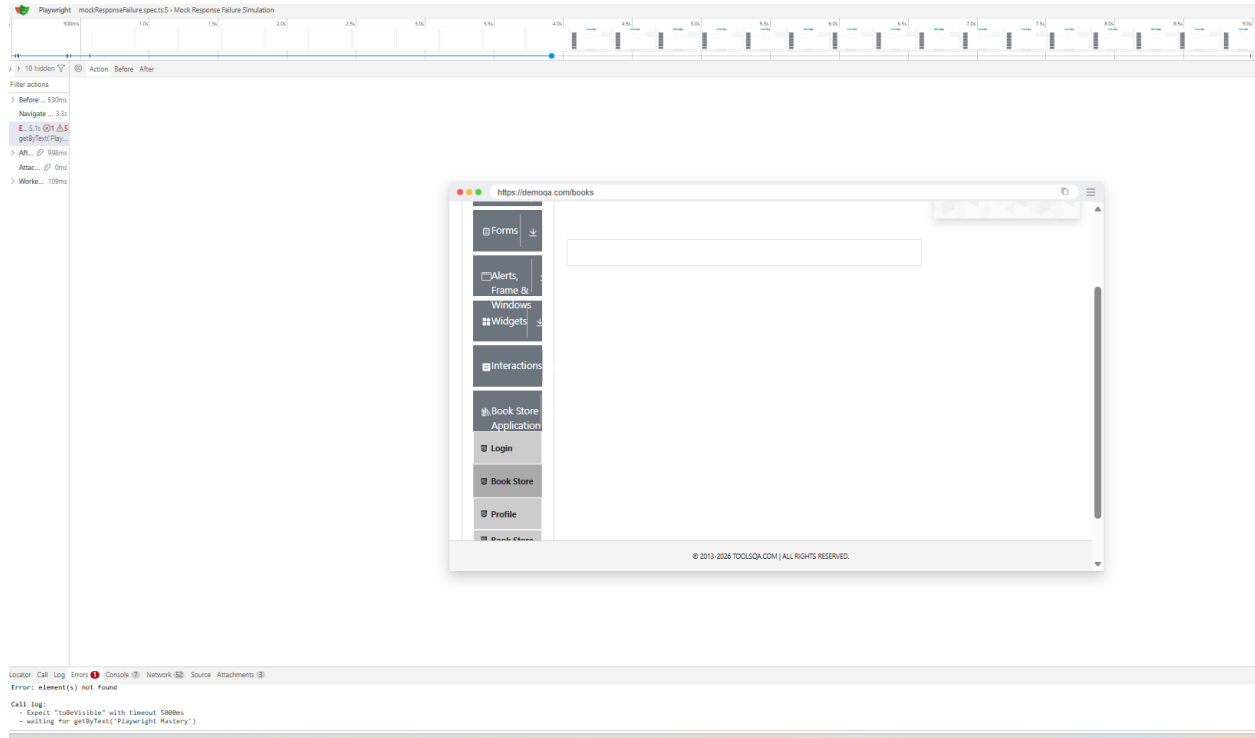
Traces



[Download](#)

Videos

TraceViewer Evidence:



Root Cause:

Mocked response schema did not match the application's expected API contract.

Resolution:

Restored correct response structure.

5) Browser Window Handling Failure: Simulate failure while handling child tabs

Steps used to simulate:

1. Create a new test file: browserWindowFailure.spec.ts
2. Add the following test:

```
import { test } from '@playwright/test';

test('Browser Window Handling Failure', async ({ page }) => {

    await page.goto('https://demoqa.com/browser-windows');

    await page.locator('#tabButton').click(); // Intentionally validating on parent
    page

    await page.locator('#sampleHeading').textContent();

});
```

3. Execute the test:

```
npx playwright test browserWindowFailure.spec.ts
```

(Changes made are: removed the childPage handling logic and attempted to validate the newly opened tab content using the parent page reference)

Failure Log:

```
PS C:\Users\dnyanesh.panchal\Desktop\AutomationTask_2> npx playwright test browserWindowFailure.spec.ts

Running 3 tests using 3 workers
1) [chromium] > tests\browserWindowFailure.spec.ts:2:5 > Browser Window Handling Failure -----

Test timeout of 30000ms exceeded.

Error: locator.textContent: Test timeout of 30000ms exceeded.
Call log:
  - waiting for locator('#sampleHeading')
```

HTML Report of the Test:

Browser Window Handling Failure

browserWindowFailure.spec.ts:2

View Trace | 33.1s

chromium

Run

Errors

Test timeout of 30000ms exceeded.

Copy prompt

Error: locator.textContent: Test timeout of 30000ms exceeded.

Call log:

- waiting for locator("#sampleHeading")

```
3 |   await page.goto('https://demoqa.com/browser-windows');
4 |   await page.locator("#tabButton").click(); // Intentionally validating on parent page
> 5 |   await page.locator("#sampleHeading").textContent();
    |                                     ^
6 | });
   at C:\Users\dnyanesh.panchal\Desktop\AutomationTask_2\tests\browserWindowFailure.spec.ts:5:42
```

Test Steps

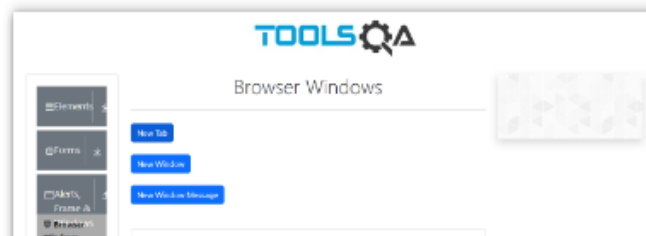
Filter steps

- > ✓ Before Hooks 981ms
- ✓ Navigate to "/browser-windows" — browserWindowFailure.spec.ts:3 5.8s
 - 2 | test("Browser Window Handling Failure", async ({ page }) => {
 - 3 | await page.goto('https://demoqa.com/browser-windows');
 - 4 | await page.locator("#tabButton").click(); // Intentionally validating on parent page
- > ✓ Click locator("#tabButton") — browserWindowFailure.spec.ts:4 453ms
- > ✓ After Hooks 3.1s
- > ✓ Worker Cleanup 104ms

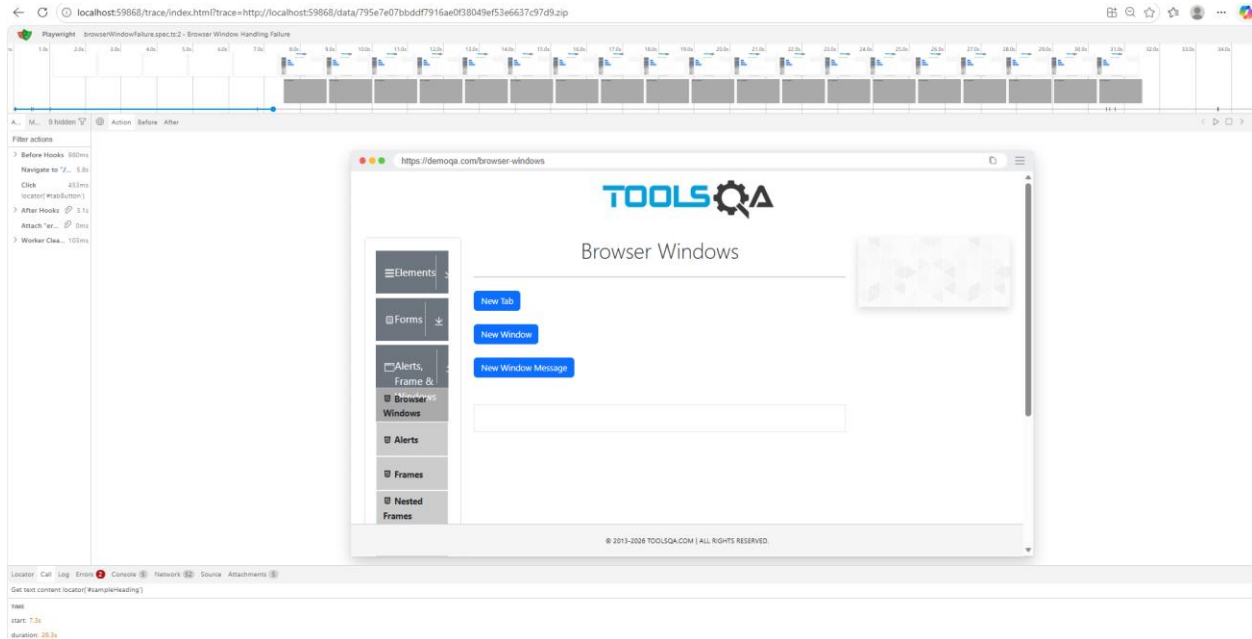
Screenshots

This is a sample page

screenshot



TraceViewer Evidence:



Root Cause:

The New tab button opens a separate browser tab containing the top element:

#sampleHeading

However, the test attempted to locate the element on the parent page instead of switching to the newly opened child tab and as a result Playwright continuously searched for the element on the wrong browser context until the timeout period was exceeded.

Resolution:

Capture the child tab and perform all validations on childPage instead of page

Timeout Types

1) Test Timeout: To simulate a test failure caused by exceeding the overall test execution timeout configured in Playwright

Steps used to simulate:

1. Create a new test file: testTimeoutFailure.spec.ts
2. Add the following test:

```
import { test } from '@playwright/test';

test('Test Timeout Simulation', async ({ page }) => {
    test.setTimeout(1000); await page.waitForTimeout(3000);
});
```

3. Execute the following:
npx playwright test testTimeoutFailure.spec.ts

Failure Log:

```
PS C:\Users\dnyanesh.panchal\Desktop\AutomationTask_2> npx playwright test testTimeoutFailure.spec.ts

Running 3 tests using 3 workers
1) [webkit] › tests\testTimeoutFailure.spec.ts:2:5 › Test Timeout Simulation

Test timeout of 1000ms exceeded.

Error: page.waitForTimeout: Test timeout of 1000ms exceeded.

   1 | import { test } from '@playwright/test';
   2 | test('Test Timeout Simulation', async ({ page }) => {
>  3 |     test.setTimeout(1000); await page.waitForTimeout(3000);
     |                               ^
   4 | });
```

HTML Report of the test:

All 1

Passed 0

✖ Failed 1

Flaky 0

Skipped 0

next >

Test Timeout Simulation

testTimeoutFailure.spec.ts:2

View Trace | 1.4s

chromium

Run

Errors

Test timeout of 1000ms exceeded.

Copy prompt

Error: page.waitForTimeout: Test timeout of 1000ms exceeded.
1 | import { test } from '@playwright/test';
2 | test('Test Timeout Simulation', async ({ page }) => {
> 3 | test.setTimeout(1000); await page.waitForTimeout(3000);
4 | });
 at C:\Users\dnyanesh.panchal\Desktop\AutomationTask_2\tests\testTimeoutFailure.spec.ts:3:39

Test Steps

Filter steps

> ✓ Before Hooks

> ✖ Wait for timeout — testTimeoutFailure.spec.ts:3

> ✓ After Hooks

> ✓ Worker Cleanup

994ms

860ms

375ms

152ms

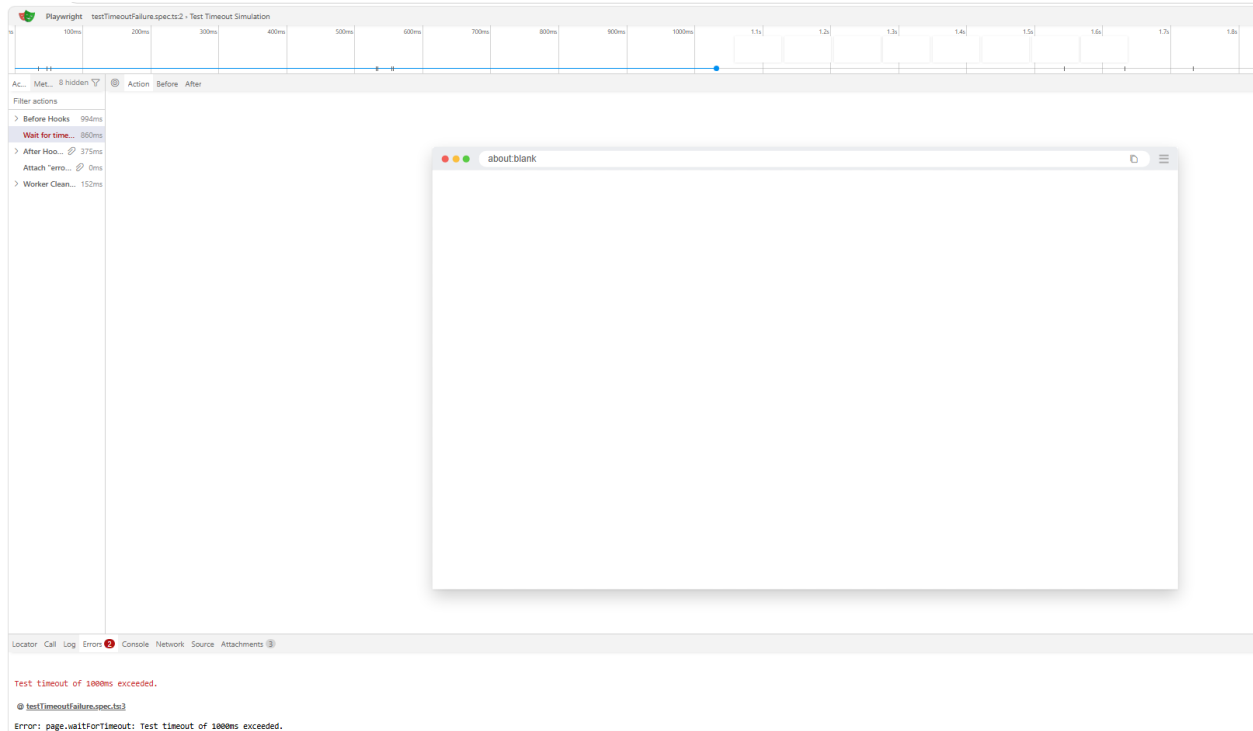
Screenshots

[screenshot](#)

Traces

[trace](#)

TraceViewer Evidence:



Root Cause:

The test timeout was explicitly configured to 1000 milliseconds while the test execution was intentionally delayed for 3000 milliseconds. Since the overall test duration exceeded the configured timeout limit, Playwright terminated the test execution.

Resolution:

Increase the test timeout value or optimize test execution time

2) Action Timeout : To simulate a timeout occurring during execution of a user action on a web element.

Steps used to simulate:

1. Create a new test file: actionTimeoutFailure.spec.ts
2. Add the following test:

```
import { test } from '@playwright/test';

test('Action Timeout Simulation', async ({ page }) => {
  await page.goto( 'https://demoqa.com/browser-windows' );
  await page.locator('#invalidButton').click({ timeout: 2000 });
});
```

3. Execute the following:

`npx playwright test actionTimeoutFailure.spec.ts`

Failure log:

```
PS C:\Users\dnyanesh.panchal\Desktop\AutomationTask_2> npx playwright test actionTimeoutFailure.spec.ts

Running 3 tests using 3 workers
1) [chromium] › tests\actionTimeoutFailure.spec.ts:2:5 › Action Timeout Simulation -----

    TimeoutError: locator.click: Timeout 2000ms exceeded.
    Call log:
      - waiting for locator('#invalidButton')

2 | test('Action Timeout Simulation', async ({ page }) => {
3 |   await page.goto( 'https://demoqa.com/browser-windows' );
```

HTML report of the test:

All 3
Passed 0
Failed 3
Flaky 0
Skipped 0

next >

Action Timeout Simulation

actionTimeoutFailure.spec.ts:2

chromium

Run

Errors

```

TimeoutError: locator.click: Timeout 2000ms exceeded.
Call log:
- waiting for locator("#invalidButton")

2 | test('Action Timeout Simulation', async ({ page }) => {
3 |   await page.goto('https://demoqa.com/browser-windows');
> 4 |   await page.locator('#invalidButton').click({ timeout: 2000 });
    |                                     ^
5 | });
   at C:\Users\dnyanesh.panchal\Desktop\AutomationTask_2\tests\actionTimeoutFailure.spec.ts:4:42

```

Copy prompt

Test Steps

> ✓ Before Hooks	840ms
> ✓ Navigate to "/browser-windows" — actionTimeoutFailure.spec.ts:3	6.1s
> ✗ Click locator("#invalidButton") — actionTimeoutFailure.spec.ts:4	2.1s
> ✓ After Hooks	1.2s
> ✓ Worker Cleanup	114ms

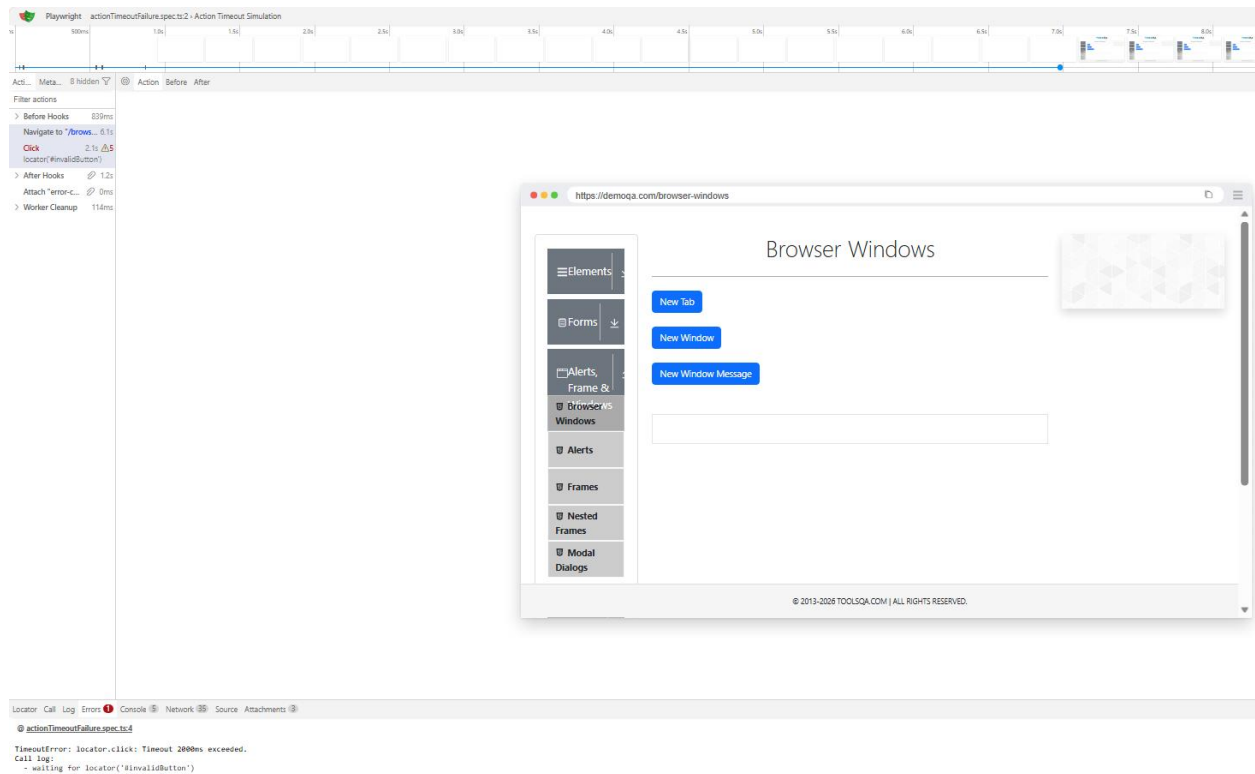
Screenshots

screenshot

Traces

trace

TraceViewer Evidence:



Root Cause:

The locator used in the click action did not exist on the page. Playwright continuously attempted to locate the element until the configured action timeout period expired.

Resolution:

Verify locator accuracy and ensure the target element exists before performing actions.

3. Expect Timeout: To simulate a timeout occurring during assertion validation

Steps used to Simulate:

1. Create a new test file: expectTimeoutFailure.ts
2. Add the following test:

```
import { test, expect } from '@playwright/test';

test('Expect Timeout Simulation', async ({ page }) => {

    await page.goto( 'https://demoqa.com/browser-windows' );

    await expect( page.getByText('Batman') ).toBeVisible({ timeout: 2000 });

});
```

3. Execute the following:
npx playwright test expectTimeoutFailure.spec.ts

Failure log:

```
❖ PS C:\Users\dnyanesh.panchal\Desktop\AutomationTask_2> npx playwright test expectTimeoutFailure.spec.ts

Running 3 tests using 3 workers
1) [webkit] > tests\expectTimeoutFailure.spec.ts:2:5 > Expect Timeout Simulation -----

Error: expect(locator).toBeVisible() failed

Locator: getByText('Batman')
Expected: visible
Timeout: 2000ms
Error: element(s) not found
```

HTML Report of the test:

All 3
Passed 0
Failed 3
Flaky 0
Skipped 0

[next >](#)

Expect Timeout Simulation

expectTimeoutFailure.spec.ts:2

chromium

Run

Errors

Error: expect(locator).toBeVisible() failed
Locator: getByText('Batman')
Expected: visible
Timeout: 2000ms
Error: element(s) not found

Call log:
- Expect "toBeVisible" with timeout 2000ms
- waiting for getByText('Batman')

Copy prompt

```

2 | test('Expect Timeout Simulation', async ({ page }) => {
3 |   await page.goto('https://demoqa.com/browser-windows');
> 4 |   await expect(page.getByText('Batman')).toBeVisible({ timeout: 2000 });
    |                                     ^
5 | });
   at C:\Users\dnyanesh.panchal\Desktop\AutomationTask_2\tests\expectTimeoutFailure.spec.ts:4:46

```

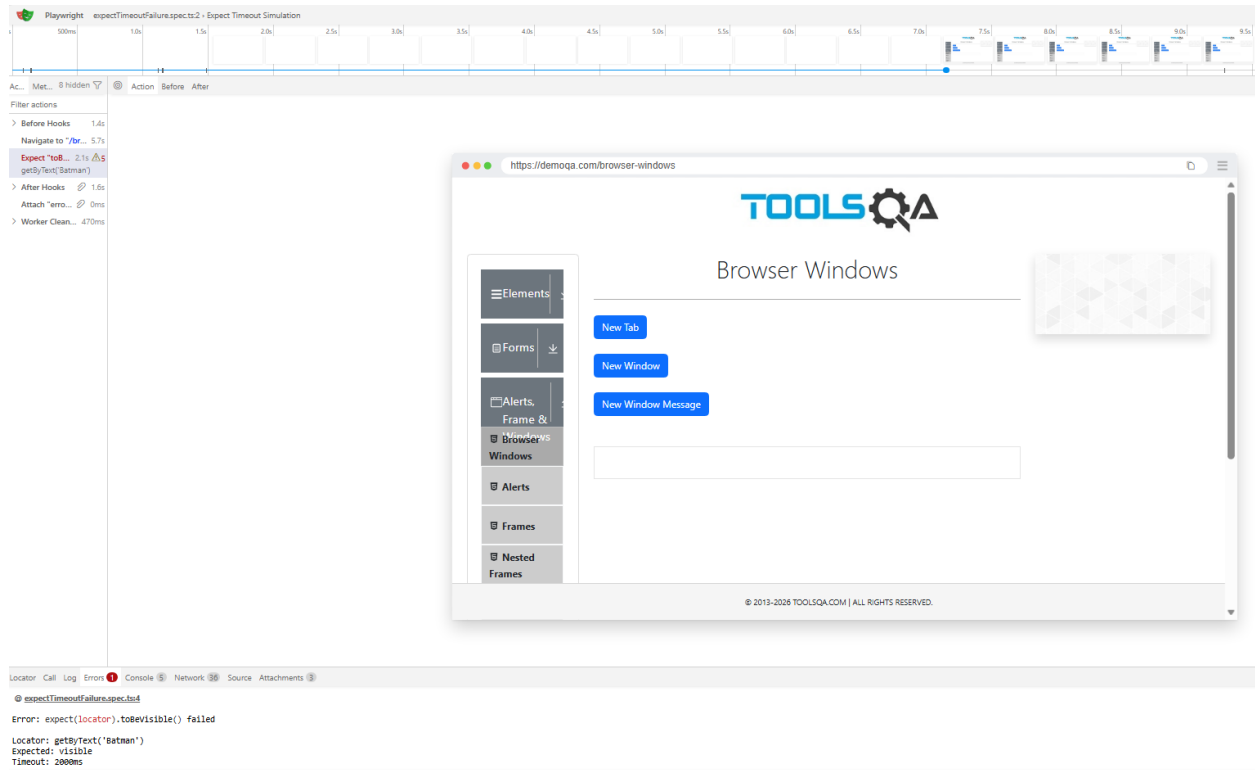
Test Steps

> ✓ Before Hooks	1.4s
> ✓ Navigate to "/browser-windows" — expectTimeoutFailure.spec.ts:3	5.7s
> ✗ Expect "toBeVisible" getByText("Batman") — expectTimeoutFailure.spec.ts:4	2.1s
> ✓ After Hooks	1.6s
> ✓ Worker Cleanup	470ms

Screenshots

[screenshot](#)

TraceViewer Evidence:



Root Cause:

The assertion expected text that never appeared on the page. Playwright repeatedly retried the assertion until the expectation timeout period was reached.

Resolution:

Verify expected values and ensure that the application displays the required content before performing assertions.